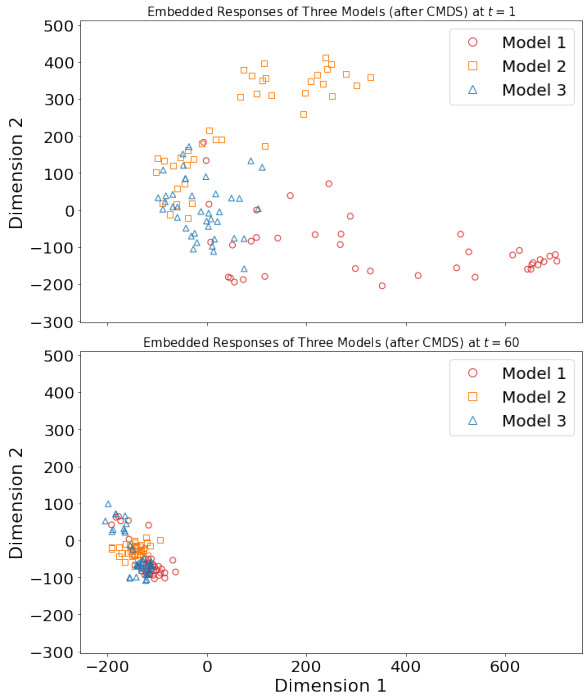

Figure 3: Scatter plots of the embedded responses from three LLMs in a 2D space at $t = 1$ and $t = T$, with small random jitter added to reduce overlap and improve visibility. At $t = 1$, the models exhibit clearly distinct response distributions. By $t = T$, the responses become tightly clustered and more similar across models.

we can further prove our claim (5) that $\|D^{(t)}\|_F$ converges to 0 in probability. We present the sketch of the proof in Appendix A.

When comparing the simulation results of our LLM and GMM experiments, we observe similar phenomena. Prior work, such as Shumailov et al. (2024), has also drawn parallels between LLM experiments and GMM setups. As a result, establishing theoretical claims in the GMM setting offers theoretical support for the behaviors observed in our LLM experiments.

However, we observe that the pattern of diminishing norms in Figure 4 looks smoother than that of LLMs in Figure 2. This is because two-component GMM is much simpler than LLM. Using GMMs with a larger number of components serves as a better proxy for LLMs, as the additional components offer greater capacity to capture complex distributions.

Figure 6 gives an example of our experiment using three GMMs of $B = 11$ components. $\left\|D^{(t)}\right\|_F$ also seems to converge to 0, but the line looks less smooth and closer to the LLM plot. Interestingly, we see a sudden fall of $\left\|D^{(t)}\right\|_F$ around $t = 160$ in Figure 6. Therefore, it is highly likely that in our

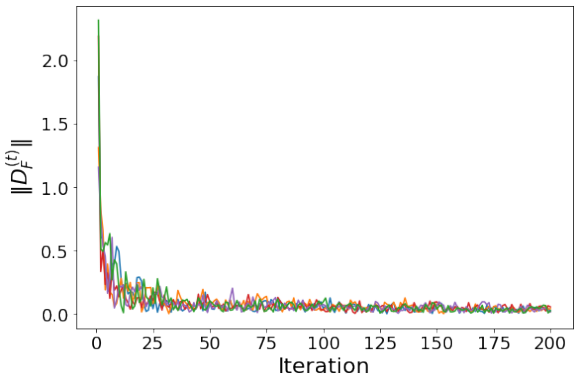

Figure 4: As t increases, the norm of the distance matrix in our GMM experiment (with $B = 2$), i.e. $\left\|D^{(t)}\right\|_F$ in Eq.(4), decreases to near zero across all 5 replicates.

LLM experiment, the norm of the distance matrix between models will continue to decrease at some larger t. This highlights another advantage of our GMM setup—it serves as a useful proxy by offering insights into potential behaviors in the LLM experiment.

5 Discussion

Our experiments demonstrate that model collapse can be observed and quantified in a fully API-accessible LLM network with RAG engine that mimics real-world Internet usage. By embedding individual responses and tracking the variation in induced Frobenius norm of the pairwise distance matrix, we observe a norm decay toward a small value, indicating that all agents have converged to an information neutral equilibrium. Importantly, this convergence should not be viewed as intrinsically *good* or *bad*; it merely signifies that further training using synthetic data shall no longer alter communication dynamics. Essentially the network used up all the information entropy from the raw data and communication built on the network is stabilized. Whether such "neutral" collapse is detrimental depends on goals of downstream tasks.

Our GMM setup within the LWD framework exhibits limiting behavior that closely mirrors that of our LLM experiment. One minor difference is the limiting value: we prove that the distance matrix norm in the GMM setting converges to zero in expectation, whereas in the LLM network, we only conjecture convergence to a small, non-negative value. The discrepancy arises because a B-component GMM is restricted to a finite parametrization, while a Transformer-based LLM occupies an effectively infinite-dimensional func-

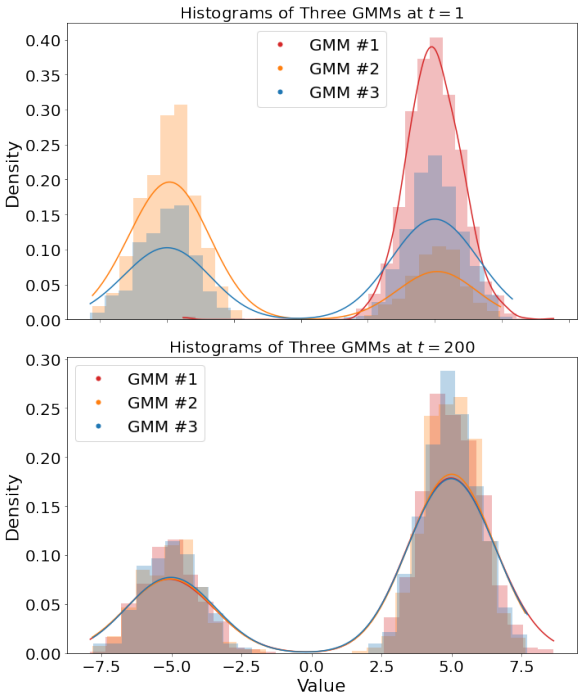

Figure 5: Histograms with density curves for three GMMs. At $t = 1$, the distributions are clearly distinct, while at $t = T$, the density curves nearly coincide.

tion space (Pérez et al., 2019; Yang, 2020).

This distinction underscores a key advantage of our GMM proxy: its extensibility via changing parameterization. As demonstrated in Section 4, GMMs with more components produce convergence patterns that more similar to those observed in real LLMs, illustrating the framework's improved fidelity over simpler statistical surrogates. Moreover, in our setup, the means and variances of the Gaussians are fixed, and only the mixture weights are updated. These parameters can also be made time-varying, providing additional flexibility in approximating LLM dynamics. Importantly, simulating interactions in such complicated GMM system requires only minutes of computation, offering a highly efficient alternative for studying synthetic data pipelines before committing to expensive LLM experiments.

6 Limitations

Despite the promise of our LLM Web Dynamics, several limitations remain. Our current study focuses on *pattern learning* where we characterize collapse by geometric drift in terms of the diminishing norm of pairwise distance matrix in the embedded vector space. Because LLMs and their interaction graphs are essentially black-box models, these results could also be better justified by *further statistical inference*. Therefore a key next step is to transform the pattern learning into systematic hypothesis testing or inference framework, providing finite-sample, high-probability guarantees. Finally, the LWD with RAG engine serves as a controllable testbed to quantify, isolate and exam the effect of different variables during the process of model collapse. Covariate like synthetic-to-real ratio, the distribution and quality of real data, and possible irrelevant noisy data perturbation can be incorporated into the framework LWD and produce predictable model collapse for various downstream tasks.

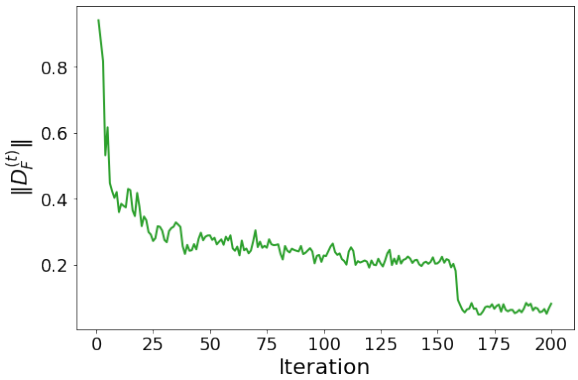

Figure 6: As t increases, the norm of the distance matrix in our GMM experiment (with $B = 11$) exhibits a decreasing trend, although the trajectory appears less smooth compared to Figure 4. Notably, the norm continues to decline at some large t.

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